

Large Language Model Agents Enabled Generative Design of Fluidic Computation Interfaces

Qiuyu Lu*

University of California, Berkeley
California, CA, USA
qiuyulu@berkeley.edu

Jiawei Fang*

University of California, Berkeley
California, CA, USA
jiaweif@berkeley.edu

Zhihao Yao*

Tsinghua University
Beijing, China
yaozh_h@outlook.com

Yue Yang

University of California, Berkeley
California, CA, USA
yue.yang@berkeley.edu

Shiqing Lyu

Tsinghua University
Beijing, China
lvsq22@mails.tsinghua.edu.cn

Haipeng Mi

Tsinghua University
Beijing, China
mhp@tsinghua.edu.cn

Lining Yao

University of California, Berkeley
California, CA, USA
liningy@berkeley.edu

ABSTRACT

The creation of interactive devices is a major area of interest. However, traditional design tools in this field often require a significant learning curve and may not effectively support creative ideation. This study explores the use of fluidic computation interfaces as a case study to examine the potential of enhancing design tools for physical devices with Large Language Model (LLM) agents. With LLM agents, the Generative Design Tool (GDT) can understand the capabilities and limitations of new devices, suggest diverse, insightful, and practical application scenarios, and recommend designs that are technically and contextually appropriate. Additionally, it generates the necessary design parameters for the traditional components of the design tool to visualize results and create files for fabrication.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Natural language processing**.

KEYWORDS

Generative Design, Large Language Model, Novel Devices

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1 INTRODUCTION

In Human-Computer Interaction (HCI), designing interactive devices is a significant endeavor. This process involves specialized tools that simplify parametric designing and provide tailored simulation and fabrication instructions [1, 5, 7, 14, 15]. However, designing a novel interactive device using existing specific design tools could be challenging, requiring a comprehensive understanding of device capabilities, limitations, and design modules. Even experts must invest significant effort in creating specific use cases that meet diverse requirements and demonstrate the technology's application value.

LLMs excel in creative endeavors due to their broad knowledge base, nuanced language generation capabilities, and contextual awareness [2–4, 6]. To explore how LLM agents can contribute to the design workflows of interactive physical devices [8–10, 12, 13], we developed a generative design tool (GDT) augmented with LLM agents for the fluidic computation interface (FCI) proposed by Lu et al. [11]. This tool addresses designer inquiries about FCI, provides design inspiration, offers detailed function suggestions, guides users through design steps, verifies final designs, and generates 3D models with interaction simulations. We discuss the advantages and challenges of integrating LLM agents into specialized design tools, aiming to inspire and benefit researchers in this field. The contribution of this work lies in exploring an approach for developing LLM-enhanced design tools for novel devices (specifically FCI) and developing a GDT for FCI that integrates LLM agents to apply fluid computation kits to new building devices.

2 OVERVIEW

The GDT's includes elements similar to those in conventional design tools, such as parametric-driven design, design outcome rendering previews, and simulations. However, unlike those tools that

*Contributed equally

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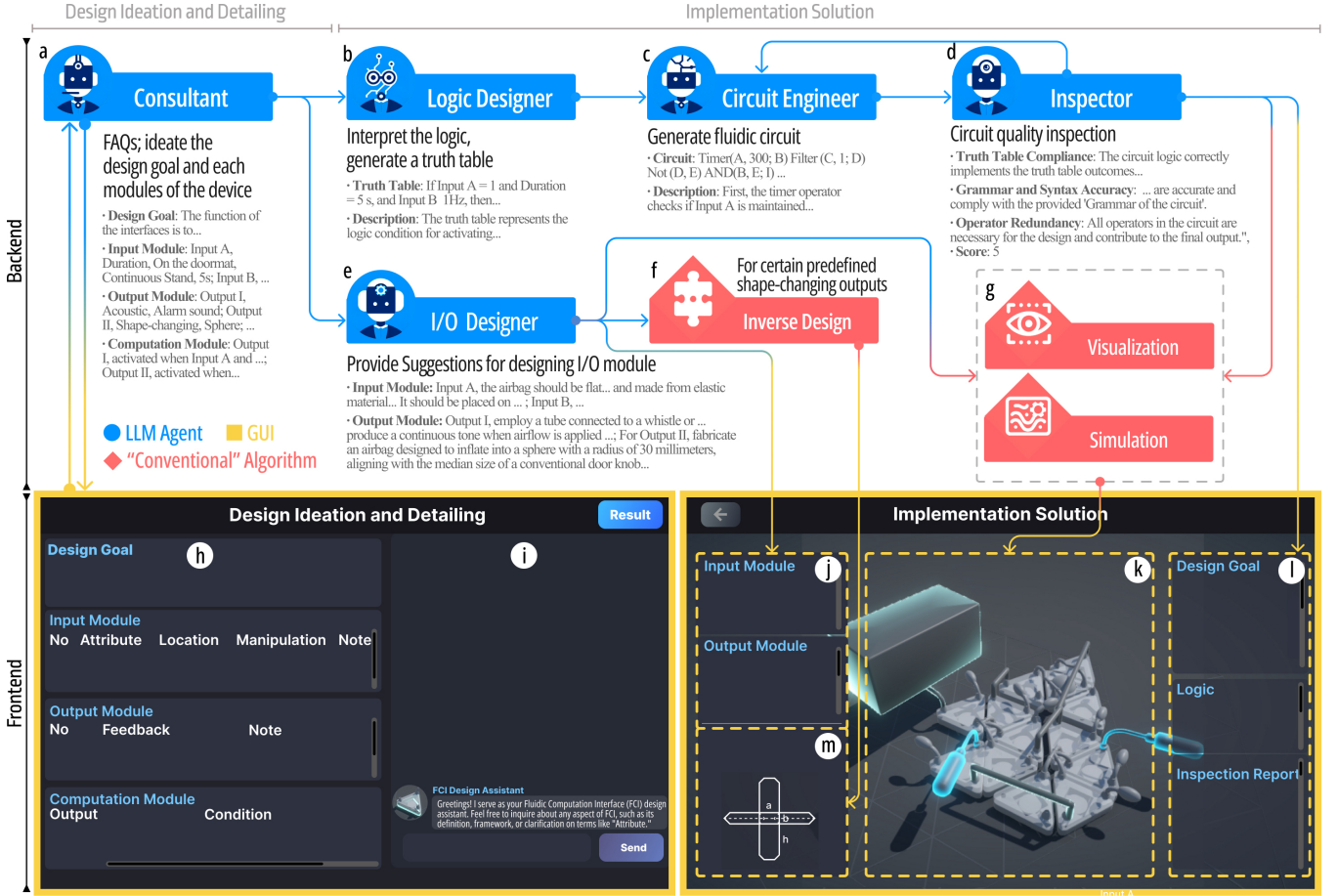


Figure 1: The GDT's overview. Sections a-g constitute the backend, which includes LLM agents and conventional algorithms. Sections h-m represent the frontend, utilized by users interacting with the GDT. The *Consultant* agent (a) and the first tab of the GUI (h,j) are dedicated to assisting with the design ideation and detailing phase. The remaining components are focused on the solution implementation phase. Some text within the GUI has been enlarged to enhance readability. The heat-sealing pattern for the output module (m) will be provided when the type of output is shape-changing, and the shape is associated with an inverse design algorithm.

require users to conceive application designs (design goals, DGs) that fit new devices and manually set various parameters or drag-and-drop modules to complete the design, this GDT, augmented with LLM agents, can assist or even take over these tasks.

3 IMPLEMENTATION OF THE GDT

Agents Architecture. The architecture utilizes OpenAI APIs (GPT-4 preview), Voiceflow, Flask, and Unity in its development. Unity serves as the primary front-end interface, while Voiceflow, Flask, and OpenAI APIs form the backbone of the back-end, enabling the collaborative workflow of LLM agents.

Configure Agents. To configure agents, we primarily assign agents to focus on narrow and specific types of work and categorize the entire design process into two main phases: ideation of design goals and detailing (Ideation & Detailing), and the technical implementation of solutions (Implementation Solution), and qualify agents for their assigned tasks through Libraries, Instruction, Tool. Besides,

we implement Standardized Information Formats for communication and ensure the agents Receive Necessary Information Only to ensure facilitate collaboration among agents

4 PILOT EVALUATION AND DISCUSSION

The GDT was tasked with proposing one hundred DGs, with results showing that it can provide very diverse DGs to facilitate the ideation, spanning ten major categories and twenty-four subcategories based on application scenarios. With twenty-seven selected DGs, the GDT can generally generate appropriate design details and parameters to support building the corresponding fluidic computation interfaces.

This study targets the growing need to improve specialized design tools for novel interactive devices, suggesting the use of LLM agents as a practical enhancement. Future work could further explore performance in the Human-AI co-design process through user study, integrating multimodality like image output, exploring the scalability of the tool, etc.

REFERENCES

- [1] Byoungkwon An, Ye Tao, Jianzhe Gu, Tingyu Cheng, Xiang 'Anthony' Chen, Xiaoxiao Zhang, Wei Zhao, Youngwook Do, Shigeo Takahashi, Hsiang-Yun Wu, Teng Zhang, and Lining Yao. 2018. Thermorph: Democratizing 4D Printing of Self-Folding Materials and Interfaces (*CHI '18*). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3173574.3173834>
- [2] Bhavya Bhavya, Jinjun Xiong, and Chengxiang Zhai. 2023. CAM: A Large Language Model-based Creative Analogy Mining Framework (*WWW '23*). Association for Computing Machinery, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3543507.3587431>
- [3] Yu Cheng, Jieshan Chen, Qing Huang, Zhenchang Xing, Xiwei Xu, and Qinghua Lu. 2023. Prompt Sapper: A LLM-Empowered Production Tool for Building AI Chains. (dec 2023). <https://doi.org/10.1145/3638247>
- [4] Peitong Duan, Jeremy Warner, and Bjoern Hartmann. 2023. Towards Generating UI Design Feedback with LLMs (*UIST '23 Adjunct*). Association for Computing Machinery, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3586182.3615810>
- [5] Emily Guan, Di Wu, Qiuyu Lu, and Lining Yao. 2024. Design and Simulation Tool for Sequentially and Conditionally Programmable Waxpaper Morphing Interfaces. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI EA '24*). Association for Computing Machinery, New York, NY, USA, Article 399, 5 pages. <https://doi.org/10.1145/3613905.3648661>
- [6] Hyung-Kwon Ko, Gwanmo Park, Hyeon Jeon, Jaemin Jo, Juho Kim, and Jinwook Seo. 2023. Large-scale Text-to-Image Generation Models for Visual Artists' Creative Works (*IUI '23*). Association for Computing Machinery, New York, NY, USA, 15 pages. <https://doi.org/10.1145/3581641.3584078>
- [7] Qiuyu Lu, Yejun Liu, and Haipeng Mi. 2020. MotionFlow: Time-axis-based Multiple Robots Expressive Motion Programming. In *Proceedings of the 3rd International Conference on Computer Science and Software Engineering* (Beijing, China) (*CSSE '20*). Association for Computing Machinery, New York, NY, USA, 145–149. <https://doi.org/10.1145/3403746.3403919>
- [8] Qiuyu Lu, Chengpeng Mao, Liyuan Wang, and Haipeng Mi. 2016. LIME: LIquid MEtal Interfaces for Non-Rigid Interaction. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology* (Tokyo, Japan) (*UIST '16*). Association for Computing Machinery, New York, NY, USA, 449–452. <https://doi.org/10.1145/2984511.2984562>
- [9] Qiuyu Lu, Jifei Ou, João Wilbert, André Haben, Haipeng Mi, and Hiroshi Ishii. 2019. milliMorph – Fluid-Driven Thin Film Shape-Change Materials for Interaction Design. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (*UIST '19*). Association for Computing Machinery, New York, NY, USA, 663–672. <https://doi.org/10.1145/3332165.3347956>
- [10] Qiuyu Lu, Jifei Ou, Lining Yao, and Hiroshi Ishii. 2024. milleCrepe: Extending Capabilities of Fluid-driven Interfaces with Multilayer Structures and Diverse Actuation Media. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems* (*CHI EA '24*). Association for Computing Machinery, New York, NY, USA, Article 383, 10 pages. <https://doi.org/10.1145/3613905.3650891>
- [11] Qiuyu Lu, Haiqing Xu, Yijie Guo, Joey Yu Wang, and Lining Yao. 2023. Fluidic Computation Kit: Towards Electronic-free Shape-changing Interfaces. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (*CHI '23*). Association for Computing Machinery, New York, NY, USA, 1–21. <https://doi.org/10.1145/3544548.3580783>
- [12] Qiuyu Lu, Tianyu Yu, Semina Yi, Yuran Ding, Haipeng Mi, and Lining Yao. 2023. Sustainflatable: Harvesting, Storing and Utilizing Ambient Energy for Pneumatic Morphing Interfaces. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (<conf-loc>, <city>San Francisco</city>, <state>CA</state>, <country>USA</country>, </conf-loc>) (*UIST '23*). Association for Computing Machinery, New York, NY, USA, Article 32, 20 pages. <https://doi.org/10.1145/3586183.3606721>
- [13] Di Wu, Emily Guan, Yunjia Zhang, Hsuanju Lai, Qiuyu Lu, and Lining Yao. 2024. Waxpaper Actuator: Sequentially and Conditionally Programmable Wax Paper for Morphing Interfaces. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, New York, NY, USA, Article 865, 16 pages. <https://doi.org/10.1145/3613904.3642373>
- [14] Humphrey Yang, Kuanren Qian, Haolin Liu, Yuxuan Yu, Jianzhe Gu, Matthew McGehee, Yongjie Jessica Zhang, and Lining Yao. 2020. SimuLearn: Fast and Accurate Simulator to Support Morphing Materials Design and Workflows. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (*UIST '20*). Association for Computing Machinery, New York, NY, USA, 71–84. <https://doi.org/10.1145/3379337.3415867>
- [15] Tianyu Yu, Mengjia Niu, Haipeng Mi, and Qiuyu Lu. 2024. MilliWare: Parametric Modeling and Simulation of Millifluidic Shape-changing Interface. In *Proceedings of the Eleventh International Symposium of Chinese CHI* (Denpasar, Bali, Indonesia) (*CHCHI '23*). Association for Computing Machinery, New York, NY, USA, 461–467. <https://doi.org/10.1145/3629606.3629654>